

ALEXANDRE AMICE

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EDUCATION

Massachusetts Institute of Technology (MIT)

August 2020 – May 2026

Doctor of Philosophy: Electrical Engineering and Computer Science

Dissertation: Strong Convex Relaxations and Tailored Conic Solvers with Applications in Robotics

Advisors: Pablo Parrilo and Russ Tedrake

University of Pennsylvania

August 2018 – May 2020

Master of Science in Engineering: Robotics

Thesis: Optimal Constraint Relaxation: Theory and Application to Problems in Robotics

Advisors: Alejandro Ribeiro and C.J. Taylor

University of Pennsylvania

August 2016 – May 2020

Bachelor of Science in Engineering: Electrical Engineering and Mathematics

Summa Cum Laude

RESEARCH INTERESTS

Optimization and robotics researcher developing numerical methods for AI and autonomy problems where speed, solution quality, and reliability all matter. Prior work spans semidefinite and sums-of-squares programming, first-order conic solvers, graph-structured optimization, and high-performance linear algebra with an emphasis on robust open-source implementations.

ACADEMIC PUBLICATIONS

(* denotes co-first authorship/equal contribution)

Full publication list: Google Scholar¹

Journals

- [1] Hongkai Dai, **Alexandre Amice***, Peter Werner, Annan Zhang, and Russ Tedrake. “Certified polyhedral decompositions of collision-free configuration space”. In: *The International Journal of Robotics Research* **Invited Paper** (2023).
- [2] Lujie Yang, Hongkai Dai, **Alexandre Amice**, and Russ Tedrake. “Approximate Optimal Controller Synthesis for Cart-Poles and Quadrotors via Sums-of-Squares”. In: *IEEE Robotics and Automation Letters* (2023).
- [3] Luiz FO Chamon, **Alexandre Amice**, and Alejandro Ribeiro. “Approximately supermodular scheduling subject to matroid constraints”. In: *IEEE Transactions on Automatic Control* 67.3 (2021).

In Preparation

- [1] **Alexandre Amice**, Bernhard P Graesdal, Paulo HS Silva, Pablo A Parrilo, and Russ Tedrake. “CCosmo: A Family of First-Order Conic Solvers”. 2026. Forthcoming manuscript. **Invited talk, MOPTA**.
- [2] **Alexandre Amice**, Bernhard P Graesdal, Russ Tedrake, and Pablo A Parrilo. “A Vertex-Edge-Graph Split for Solving Large-Scale Graph of Convex Sets Problems”. 2026. Forthcoming manuscript.
- [3] **Alexandre Amice**, Bernhard P Graesdal, Russ Tedrake, and Pablo A Parrilo. “Low-Degree Implied Equalities for Strengthening Semidefinite Relaxations”. 2026. Forthcoming manuscript. **Spotlight, ICRA Frontiers of Optimization Workshop**.

Conferences

- [1] Savva Morozov, Tobia Marcucci, Bernhard Paus Graesdal, **Alexandre Amice**, Pablo A Parrilo, and Russ Tedrake. “Mixed discrete and continuous planning using shortest walks in graphs of convex sets”. In: *Workshop on the Algorithmic Foundations of Robotics (WAFR)*. 2026.

¹<https://scholar.google.com/citations?user=eYY2nMYAAAAJ>

- [2] **Alexandre Amice**, Peter Werner, and Russ Tedrake. “Certifying Bimanual RRT Motion Plans in a Second”. In: *International Conference on Robotics and Automation (ICRA)*. 2024. **Featured in MIT News**.
- [3] Bernhard P Graesdal, Shao YC Chia, Tobia Marcucci, Savva Morozov, **Alexandre Amice**, Pablo A Parrilo, and Russ Tedrake. “Towards Tight Convex Relaxations for Contact-Rich Manipulation”. In: *Robotics: Science and Systems (RSS)*. 2024.
- [4] Savva Morozov, Tobia Marcucci, **Alexandre Amice**, Bernhard P Graesdal, Rohan Bosworth, Pablo A Parrilo, and Russ Tedrake. “Multi-Query Shortest-Path Problem in Graphs of Convex Sets”. In: *Workshop on the Algorithmic Foundations of Robotics (WAFR)*. 2024.
- [5] Peter Werner, **Alexandre Amice**, Tobia Marcucci, Russ Tedrake, and Daniela Rus. “Approximating Robot Configuration Spaces with few Convex Sets using Clique Covers of Visibility Graphs”. In: *International Conference on Robotics and Automation*. 2024.
- [6] **Alexandre Amice** and Pablo Parrilo. “Solving Least Squares Problems on Partially Ordered Sets”. In: *IEEE 60th Conference on Decision and Control (CDC)*. IEEE. 2022.
- [7] **Alexandre Amice***, Hongkai Dai*, Peter Werner, Annan Zhang, and Russ Tedrake. “Finding and Optimizing Certified, Collision-Free Regions in Configuration Space for Robot Manipulators”. In: *Workshop on the Algorithmic Foundations of Robotics (WAFR)* **Outstanding Paper Award**. 2022.
- [8] Lujie Yang, Kaiqing Zhang, **Alexandre Amice**, Yunzhu Li, and Russ Tedrake. “Discrete Approximate Information States in Partially Observable Environments”. In: *2022 American Control Conference (ACC)*. IEEE. 2022, pp. 1406–1413.
- [9] Luiz FO Chamon, **Alexandre Amice**, Santiago Paternain, and Alejandro Ribeiro. “Resilient control: Compromising to adapt”. In: *59th IEEE Conference on Decision and Control (CDC)* **Roberto Tempe Award Nominee**. IEEE. 2020.
- [10] Luiz FO Chamon, **Alexandre Amice**, and Alejandro Ribeiro. “Matroid-constrained approximately supermodular optimization for near-optimal actuator scheduling”. In: *IEEE 58th Conference on Decision and Control (CDC)*. IEEE. 2019.

Preprints

- [1] Thomas Cohn, Lihan Tang, **Alexandre Amice**, and Russ Tedrake. “A Framework for Combining Optimization-Based and Analytic Inverse Kinematics”. In: *arXiv:2602.05092* (2026).

EXPERIENCE

MIT CSAIL and LIDS: Research Assistant

August 2020 – Present

- Designed algorithms for succinctly decomposing robot configuration spaces into convex sets. Leveraged these descriptions to generate advances in motion planning algorithms for contact-rich systems.
- Contributed to Sums-of-Squares (SOS)/Semidefinite programming (SDP) formulations of motion planning through contact, optimal feedback control, and provably collision-free motion planning algorithms. Implemented formulations directly in C++ to enable realistic deployment on systems.
- Research on the exploitation of structure to accelerate solving mathematical programs arising in the context of control theory and robotic applications.
- Mentored six undergraduate and master’s researchers on optimization, planning, and robotics projects.

Robotics and AI Institute: Research Intern

June 2023 – August 2023

- Analyzed performance-robustness tradeoffs in tracking and stabilization controllers for nonlinear vehicle dynamics on the Ultra-Mobility Vehicle team.

Alelab, University of Pennsylvania: Research Assistant

Jan 2018 – July 2020

- Research in discrete and convex optimization with applications in optimal control, multi-agent control, machine learning, and computer vision.
- Completed master’s thesis on optimal trade-offs in overspecified robotics problems. Applied novel theory to credibility assignment problems in machine learning and resilient control in teams of drones.

- Designed and implemented novel radar tracking and sensor-fusion algorithms, reducing the time from first object detection to full-state estimation by a factor of six, enabling radar-only instance segmentation for the first time, and reducing computational overhead by 20%.

SOFTWARE

Drake

- Core contributor to Drake's optimization, geometry, motion planning, and symbolic algebra libraries, with 78 upstream commits across 414 files and 20k+ lines of C++/Python code.

CCosmo

- ADMM-based C++ conic solver family for standard-form, matrix-valued, and batched small dense optimization problems.

VEGA

- Decomposition-based C++ solver for very large-scale Graph of Convex Sets optimization programs.

RESEARCH MENTORSHIP

- **Audrey Lee**, MIT undergraduate.
- **Michael Zheng**, MIT undergraduate; now at Sunday Robotics.
- **Abhinav Agarwal**, Princeton undergraduate; now PhD student at MIT.
- **Paulo Henrique dos Santos Silva**, MIT undergraduate.
- **Ria Sonecha**, MIT master's student; now at Applied Intuition.
- **Charles Van Hees**, UC Louvain master's student.

SERVICE AND LEADERSHIP

Workshop Organizing

- Organizer, NeurIPS 2025 Workshop on GPU-Accelerated and Scalable Optimization (ScaleOpt).

Reviewing

- Reviewed for ICRA; RA-L; T-RO; Automatica; ICML; NeurIPS; Journal of Numerical Algebra, Control, and Optimization; and CDC.

Graduate Application Assistance Program: Mentor

August 2020 - 2023

- MIT: Worked with eight prospective graduate students from underrepresented backgrounds in STEM to strengthen their graduate and fellowship application packages.
- University of Pennsylvania: Launched a graduate application assistance program at the University of Pennsylvania to match prospective graduate students with young alumni to provide mentorship throughout the graduate application process.

TEACHING

Applied Convex Optimization: MIT 6.S098

IAP 2022

Lecturer

Algebraic Techniques and Semidefinite Optimization: MIT 6.7230

Teaching Assistant and Guest Lecturer

Spring 2023, Spring 2026

Supervisor: Pablo Parrilo

Underactuated Robotics: MIT 6.832

Teaching Assistant

Spring 2022

Supervisor: Russ Tedrake

Modern Convex Optimization: University of Pennsylvania ESE 605

Teaching Assistant

Spring 2020

Supervisor: Nikolai Matni

Control of Autonomous Systems: University of Pennsylvania ESE 421

Teaching Assistant

Fall 2019

Supervisor: Bruce Kothmann